

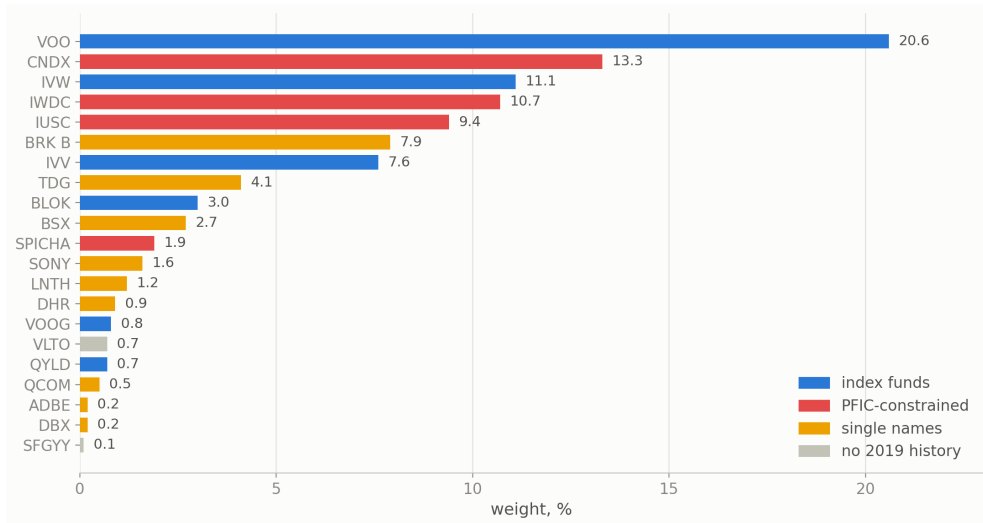
Portfolio Construction: Marchenko–Pastur Denoising & Bayesian Mean Shrinkage

Ilia Demskii

3 August 2026

Situation

CHF 574k portfolio. Reduce risk, keep covering a 0.85 % Swiss mortgage.



Marchenko–Pastur Denoised Markowitz

Focus: the covariance matrix is estimated, not known

Random-matrix theory separates resolvable structure from sampling noise

Sample correlation of CHF returns

$$E = \frac{1}{T} \tilde{X} \tilde{X}^\top, \quad q = \frac{N}{T}$$

clones removed first:

$$\rho_{ij} > 0.98$$

Split the spectrum at the fitted edge

$$\sigma_k^2 = \frac{N - \sum_{i \leq k} \lambda_i}{N - k}$$

$$\lambda_{\pm} = \sigma_k^2 \left(1 \pm \sqrt{\frac{N-k}{T}} \right)^2$$

$$\lambda_j \leq \lambda_+ \Rightarrow \lambda_j \leftarrow \bar{\lambda}_{\text{bulk}}$$

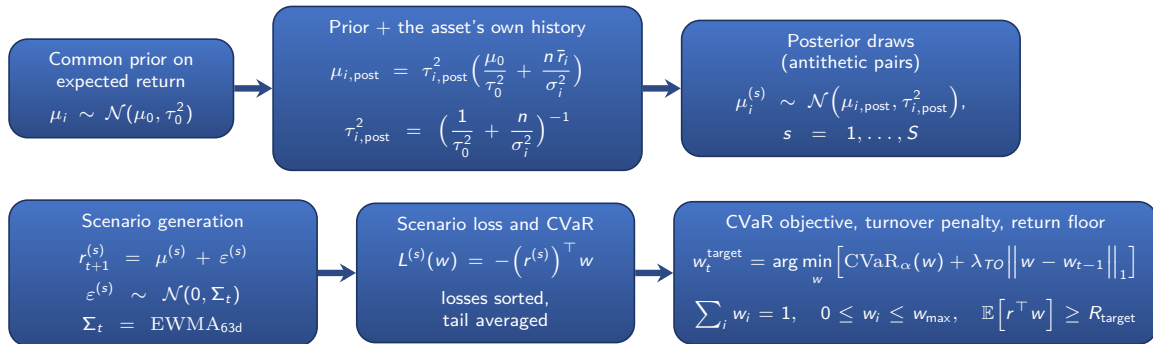
Perform return-variance optimization

$$\max_w \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma' w}}$$
$$\mathbf{1}^\top w = 1, \quad 0 \leq w \leq w_{\max}$$

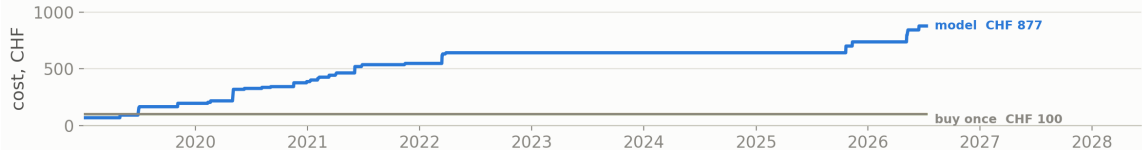
Bayesian Shrinkage Mean Model

Focus: each asset's expected return is treated as uncertain

A common prior pulls noisy per-asset means toward a shared benchmark



CHF 100k, walk-forward 2019–2026



Out-of-sample metrics, predictive diagnostics

	CAGR %	Vol %	Sharpe	VaR ₉₅ %	CVaR ₉₅ %	MaxDD %
Initial	15.09	18.43	0.82	1.75	2.76	32.77
MP fixed	12.72	16.32	0.78	1.46	2.48	30.81
Bayesian 12 %	11.76	11.37	1.03	1.03	1.68	21.18
Bayesian 14 %	11.94	12.50	0.96	1.16	1.87	23.11
S&P 500 (CHF)	14.57	20.97	0.70	1.94	3.12	33.85

